

Single-Variable Calculus

Change, Accumulation, and Approximation

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How to Read This Book

The tag system

Suggested course paths

Part I

The Two Questions of Calculus

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Core

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1.1.2 Velocity as changing distance

1.1.3 Units: miles, hours, miles per hour

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1.3.3 Area below the time axis

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Bridge

1.6 A first map of the book**1.6.1 Derivative: from distance to velocity****1.6.2 Integral: from velocity to distance****1.6.3 Approximation: how to work when exact formulas fail****1.6.4 Differential equations: when the law gives the derivative****1.7 Discovery problems and projects**

Project

Project

Bridge

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Project

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Core

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Core

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Core

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Core / Bridge

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What Derivatives Reveal

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Core / Proof

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Core

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7.1.3 Geometry problems: ladders, shadows, cones

7.1.4 Units as a guardrail

7.2 Optimization problems

Core

7.2.1 Translating a word problem into a function

7.2.2 Constraints in one variable

7.2.3 Geometry: boxes, fences, cans, and light paths

7.2.4 Endpoint checks and physical feasibility

7.3 Linear approximation in applications

Core

7.3.1 Estimating hard numbers quickly

7.3.2 Error propagation

7.3.3 Relative and percentage error

7.3.4 Sensitivity in measurement

7.4 Newton's method

Core / Computing

7.4.1 Tangent lines as root finders

7.4.2 The iteration formula

7.4.3 Convergence when the starting point is good

7.4.4 Failure modes: cycles and bad tangents

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Core

7.5.1 The differential equation $y' = ky$

7.5.2 Half-life and doubling time

7.5.3 Continuously compounded interest

7.5.4 Newton's law of cooling

Core / Bridge

7.6 Logistic growth and limited resources

7.6.1 Why pure exponential growth cannot last

7.6.2 The logistic differential equation

7.6.3 Equilibria and stability from a graph

7.6.4 Population and spread models

Optional / Project

7.7 Light, time, and extremal principles

7.7.1 Reflection from shortest path

7.7.2 Refraction and Snell's law

7.7.3 The seed of the calculus of variations

Counterexample

7.8 Warning examples

7.8.1 A critical point outside the physical domain

7.8.2 A local maximum that is not the absolute maximum

7.8.3 Newton's method converges to the wrong root or does not converge

7.9 Chapter review and applications

Project

7.9.1 Skills: related rates and optimization

Computing /
Project

7.9.2 Project: design a least-material container

7.9.3 Project: model cooling data

7.9.4 Hook: derivatives measure change; now we need to add change back up

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Integrals and the Fundamental Theorem

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Core

8.1.1 Constant velocity revisited

8.1.2 Piecewise constant velocity

8.1.3 Estimating distance from sampled speeds

8.1.4 Signed displacement versus total distance

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Core

8.2.1 Rectangular estimates

8.2.2 Left, right, and midpoint sums

8.2.3 Upper and lower sums

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Core

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8.3.3 Sums of powers

8.3.4 From finite sums to integrals

8.4 The definite integral

Core

8.4.1 Partitions and sample points

8.4.2 Riemann sums

8.4.3 Definition of the definite integral

8.4.4 Integrability, informally first

8.5 Signed area and net change

Core

8.5.1 Area above the axis

8.5.2 Area below the axis

8.5.3 Cancellation and meaning

8.5.4 Total accumulation from absolute value

Core

8.6 Properties of the integral

8.6.1 Linearity

8.6.2 Additivity over intervals

8.6.3 Order properties

8.6.4 Average value of a function

8.6.5 Symmetry

Proof / Optional

8.7 Existence of integrals

8.7.1 Continuous functions are integrable

8.7.2 Monotone functions are integrable

8.7.3 Functions with finitely many jump discontinuities

8.7.4 A bounded function that is not Riemann integrable

Counterexample

8.8 Chapter review and discovery problems

8.8.1 Skills: set up Riemann sums

8.8.2 Project: estimate energy use from power data

8.8.3 Computing: compare left, right, midpoint, and trapezoid estimates

8.8.4 Hook: an accumulation function has a slope, and that slope is surprisingly simple

Project

Computing

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9.1 Accumulation functions

Core

9.1.1 Area with a moving endpoint

9.1.2 $A(x) = \int_a^x f(t) dt$

9.1.3 Estimating the change in an accumulation function

9.1.4 The slope of accumulated area

9.2 Fundamental Theorem, Part I

Core / Proof

9.2.1 The derivative of an accumulation function

9.2.2 Continuity as the needed hypothesis

9.2.3 Geometric proof idea

9.2.4 Examples with variable upper limits

9.3 Fundamental Theorem, Part II

Core

9.3.1 Antiderivatives evaluate definite integrals

9.3.2 Net change theorem

9.3.3 Position from velocity

9.3.4 Accumulated rate in science and economics

9.4 Indefinite integrals

Core

9.4.1 Antiderivative families

9.4.2 The constant of integration

9.4.3 Initial value problems

9.4.4 Basic antiderivative formulas

9.5 Substitution

Core

9.5.1 Reversing the chain rule**9.5.2 Changing variables in a definite integral****9.5.3 Orientation and reversed limits****9.5.4 Substitution in motion and geometry**

Bridge

9.6 Differential notation and oriented length**9.6.1 dx as an oriented step****9.6.2 $f(x) dx$ as the thing being accumulated****9.6.3 Pulling an integrand through $x = g(u)$, without abstraction****9.6.4 The one-dimensional seed of differential forms**

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9.7 Warning examples**9.7.1 A discontinuity and the derivative of an accumulation function****9.7.2 Forgetting the new limits in substitution****9.7.3 Treating $\int f(x) dx$ like multiplication****9.7.4 Confusing area with net signed area****9.8 Chapter review and discovery problems**

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Project

9.8.2 Proof practice: FTC Part I in a simple case**9.8.3 Project: reconstruct a function from its rate****9.8.4 Hook: once integration is computable, it can measure more than area**

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10.1 Area between curves

Core

10.1.1 Top minus bottom

10.1.2 Right minus left

10.1.3 Splitting at intersections

10.1.4 Area from data and graphs

10.2 Volumes by slicing

Core

10.2.1 Cross-sectional area as a function

10.2.2 Solids of revolution: disks and washers

10.2.3 Non-circular cross sections

10.2.4 Choosing the slicing direction

10.3 Cylindrical shells

Core

10.3.1 A thin shell unwrapped

10.3.2 Shell method formula

10.3.3 When shells beat washers

10.3.4 Comparing two methods on the same solid

10.4 Mass, density, and center of mass

Core

10.4.1 Linear density and mass of a rod

10.4.2 Moments

10.4.3 Center of mass on a line

10.4.4 Lamina preview by slices

Bridge

10.5 Work and energy

Core

10.5.1 Constant force and variable force

10.5.2 Springs and Hooke's law

10.5.3 Lifting a chain

10.5.4 Pumping fluid from a tank

Core / Optional

10.6 Hydrostatic force

10.6.1 Pressure depends on depth

10.6.2 Force on a vertical plate

10.6.3 Choosing strips

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Core

10.7 Average value and probability

10.7.1 Average value of a continuous function

10.7.2 Probability density

10.7.3 Expected value

10.7.4 Normal distribution as a preview of non-elementary integrals

Core

10.8 Arc length and surface area of revolution

10.8.1 Arc length from small straight pieces

10.8.2 Arc length of a graph

10.8.3 Surface area by rotating a curve

10.8.4 Why some natural lengths have no elementary antiderivative

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Project

10.9.1 Skills: choose the small piece

Project

10.9.2 Project: compare volume methods

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10.9.4 Hook: many integrals cannot be found by substitution alone

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11.1 Integration by parts

Core

11.1.1 Reversing the product rule

11.1.2 Choosing u and dv

11.1.3 Repeated integration by parts

11.1.4 Definite integrals and boundary terms

11.2 Trigonometric integrals

Core

11.2.1 Powers of sine and cosine

11.2.2 Powers of tangent and secant

11.2.3 Product-to-sum identities

11.2.4 Strategy over memorization

11.3 Trigonometric substitution

Core

11.3.1 Square roots from circles and triangles

11.3.2 $a^2 - x^2$, $a^2 + x^2$, and $x^2 - a^2$

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11.4 Rational functions and partial fractions

Core

11.4.1 Polynomial division

11.4.2 Distinct linear factors

11.4.3 Repeated linear factors

11.4.4 Irreducible quadratic factors

11.4.5 Completing the square

11.5 Strategy for integration

Core

11.5.1 Recognizing the structure of an integrand

11.5.2 Substitution first or parts first?

11.5.3 Algebraic simplification

11.5.4 When no elementary antiderivative exists

Computing

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11.6.1 Checking without surrendering thought

11.6.2 Equivalent antiderivatives can look different

11.6.3 Branch and domain issues

11.6.4 Special functions born from integrals

Core

11.7 Improper integrals

11.7.1 Infinite intervals

11.7.2 Infinite discontinuities

11.7.3 Convergence and divergence

11.7.4 Comparison tests for improper integrals

11.7.5 The p -integrals

Counterexample

11.8 Warning examples

11.8.1 An antiderivative exists formally but the improper integral diverges

11.8.2 Cancellation hides divergence

11.8.3 A CAS gives an antiderivative with the wrong domain

11.8.4 A trigonometric substitution triangle used outside its range

11.9 Chapter review and discovery problems

Project

11.9.1 Skills: choose and apply integration techniques

Computing /
Project

11.9.2 Project: build an integration decision tree

11.9.3 Project: compare human and CAS antiderivatives

11.9.4 Hook: exact integration is only one way to compute an accumulated amount

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Numerical Integration, Error, and Computation

12.1 Why approximate integrals?

Core / Computing

12.1.1 Data instead of formulas

12.1.2 Non-elementary antiderivatives

12.1.3 Speed versus accuracy

12.1.4 The role of error estimates

12.2 Midpoint and trapezoidal rules

Core

12.2.1 Geometry of each rule

12.2.2 Composite rules

12.2.3 Error patterns from concavity

12.2.4 Error bounds

12.3 Simpson's Rule

Core

12.3.1 Quadratic approximation on pairs of intervals

12.3.2 Composite Simpson's Rule

12.3.3 Why Simpson's Rule is often much better

12.3.4 Simpson error bound

12.4 Numerical differentiation

Computing /
Optional

12.4.1 Forward, backward, and centered differences

12.4.2 Truncation error

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12.4.4 Why smaller h is not always better

12.5 Newton's method revisited

Computing

12.5.1 Stopping criteria**12.5.2 Error estimates**

Optional

12.5.3 Multiple roots**12.5.4 A first glimpse of chaotic iteration**

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12.6 Computing experiments**12.6.1 Approximate π by integration****12.6.2 Estimate a probability from a density****12.6.3 Compare rules on smooth and rough functions****12.6.4 Build a small numerical-integration program****12.7 Chapter review and discovery problems****12.7.1 Skills: numerical rules and error bounds****12.7.2 Concept check: exact value versus approximation****12.7.3 Hook: a differential equation often cannot be solved exactly either**

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Differential Equations: Laws Written as Derivatives

13.1 Modeling with differential equations

Core

13.1.1 A law that gives a rate

13.1.2 Solution as an unknown function

13.1.3 Initial conditions

13.1.4 Checking a proposed solution

13.2 Direction fields

Core

13.2.1 Slope field as a picture of a differential equation

13.2.2 Equilibrium solutions

13.2.3 Qualitative behavior

13.2.4 Reading long-term behavior from the field

13.3 Euler's method

Core / Computing

13.3.1 Following the slope field step by step

13.3.2 Step size and accumulated error

13.3.3 Improved Euler method

13.3.4 Numerical solutions as approximate functions

13.4 Separable equations

Core

13.4.1 Separating variables

13.4.2 Growth and decay revisited

13.4.3 Cooling and mixing

13.4.4 Implicit solutions

13.5 Logistic and threshold models

Core

13.5.1 Carrying capacity**13.5.2 Phase line**

Optional

13.5.3 Stable and unstable equilibria**13.5.4 Harvesting and tipping points**

Core

13.6 First-order linear equations**13.6.1 Standard form****13.6.2 Integrating factors****13.6.3 Mixing and finance examples****13.6.4 Why the integrating factor works**

Optional / Bridge

13.7 Second-order equations and vibration**13.7.1 Simple harmonic motion****13.7.2 The equation $y'' + \omega^2 y = 0$** **13.7.3 Damping****13.7.4 Driven oscillation and resonance**

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Counterexample

13.9 Warning examples**13.9.1 A solution lost by dividing by zero****13.9.2 Euler's method follows the wrong behavior with a large step****13.9.3 Same differential equation, different initial conditions****13.10 Chapter review and applications**

Project

13.10.1 Skills: solve and interpret first-order equationsComputing /
Project**13.10.2 Project: model a cooling cup of coffee****13.10.3 Project: compare Euler approximations with exact solutions****13.10.4 Hook: some curves are easier to describe by how they are traced than by $y = f(x)$**

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14.1 Parametric curves

Core

14.1.1 A curve as a moving point

14.1.2 Eliminating the parameter when possible

14.1.3 Different parametrizations of the same curve

14.1.4 Orientation and speed along a curve

14.2 Calculus with parametric curves

Core

14.2.1 Slope dy/dx from dx/dt and dy/dt

14.2.2 Horizontal and vertical tangents

14.2.3 Concavity

14.2.4 Motion in the plane as a bridge to vectors

14.3 Length and area for parametric curves

Core

14.3.1 Arc length from speed

14.3.2 Surface area from a parametrized curve of revolution

14.3.3 Area enclosed by a parametric curve

14.3.4 Cycloids and rolling motion

14.4 Polar coordinates

Core

14.4.1 Distance and angle as coordinates

14.4.2 Converting between polar and Cartesian coordinates

14.4.3 Polar graphs

14.4.4 Circles, spirals, roses, and limacons

14.5 Calculus in polar coordinates

Core

Core

14.5.1 Slope of a polar curve**14.5.2 Area in polar coordinates****14.5.3 Arc length in polar coordinates****14.5.4 Intersections and repeated tracing****14.6 Conic sections****14.6.1 Parabolas, ellipses, and hyperbolas from focus and directrix****14.6.2 Standard equations****14.6.3 Polar equations of conics****14.6.4 Eccentricity**

Optional / Project

14.7 Planetary motion as a capstone**14.7.1 Ellipses in polar form****14.7.2 Kepler's second law as swept area****14.7.3 Central force preview****14.7.4 Why this belongs to multivariable calculus next**

Bridge / Optional

14.8 Complex numbers and polar form**14.8.1 Complex numbers as points in the plane****14.8.2 Multiplication as scaling and rotation****14.8.3 Euler's formula preview****14.8.4 Oscillation written exponentially**

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14.9 Warning examples**14.9.1 A parametrization stops even when the curve continues****14.9.2 A polar curve traced twice****14.9.3 A tangent formula fails when both derivatives vanish****14.10 Chapter review and projects**Computing /
Project
Project**14.10.1 Skills: parametric and polar calculus****14.10.2 Project: design a cycloid animation****14.10.3 Project: compare polar and Cartesian descriptions of conics****14.10.4 Hook: infinite processes return, now as sums rather than areas**

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Core

15.1.1 Lists with a rule

15.1.2 Convergence and divergence

15.1.3 Monotone bounded sequences

15.1.4 Recursive sequences and fixed points

15.2 Infinite series

Core

15.2.1 Partial sums

15.2.2 Series convergence as a sequence problem

15.2.3 Geometric series

15.2.4 Telescoping series

15.3 The harmonic series and first warnings

Core

15.3.1 Terms go to zero but the series diverges

15.3.2 Necessary condition for convergence

15.3.3 The harmonic series

15.3.4 Grouping terms to see divergence

15.4 Positive series tests

Core

15.4.1 Integral test

15.4.2 Remainder estimates from the integral test

15.4.3 Direct comparison

15.4.4 Limit comparison

15.4.5 p -series

15.5 Alternating series

Core

15.5.1 Alternating Series Test**15.5.2 Alternating Series Estimation Theorem****15.5.3 Conditional convergence****15.5.4 Error as the first omitted term**

Core

15.6 Absolute convergence and stronger tests**15.6.1 Absolute versus conditional convergence****15.6.2 Ratio Test****15.6.3 Root Test****15.6.4 Strategy for testing series**

Optional / Proof

15.7 Rearrangement and the danger of infinity**15.7.1 Finite sums can be rearranged freely****15.7.2 Infinite conditionally convergent sums cannot****15.7.3 Riemann rearrangement idea****15.7.4 Why absolute convergence is safer**

Counterexample

15.8 Warning examples**15.8.1 Terms tend to zero but the series diverges****15.8.2 Ratio Test gives no information****15.8.3 Integral Test used on a nonpositive function****15.8.4 Alternating Series Test used without decreasing terms****15.9 Chapter review and discovery problems**

Project

15.9.1 Skills: choose series tests

Computing

15.9.2 Project: estimate sums with rigorous error bounds**15.9.3 Computing: numerical partial sums can mislead****15.9.4 Hook: some infinite sums are functions, not just numbers**

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Power Series, Taylor Polynomials, and Taylor Series

16.1 Power series

Core

16.1.1 A polynomial with infinitely many terms

16.1.2 Center of a power series

16.1.3 Radius and interval of convergence

16.1.4 Endpoint testing

Core

16.2 Working with power series

16.2.1 Algebra of power series

16.2.2 Differentiating power series

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16.2.4 Why power series behave better than arbitrary series

Core

16.3 Representing functions by power series

16.3.1 Geometric series as the seed

16.3.2 Series for $1/(1-x)$

16.3.3 Series for $\ln(1+x)$

16.3.4 Series for $\arctan x$

16.3.5 Estimating π from a series

Core

16.4 Taylor polynomials

16.4.1 Matching value and derivatives

16.4.2 Linear, quadratic, and cubic approximations

16.4.3 Taylor polynomial centered at a

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Core / Proof

16.5 Taylor's theorem and remainders

16.5.1 The error term matters

Optional

16.5.2 Lagrange form of the remainder**16.5.3 Integral form of the remainder****16.5.4 Alternating-series remainders as a special case**

Core

16.6 Taylor series for elementary functions**16.6.1 e^x** **16.6.2 $\sin x$ and $\cos x$** **16.6.3 $\ln(1+x)$** **16.6.4 $\arctan x$** **16.6.5 Binomial series**

Core

16.7 Applications of Taylor series**16.7.1 Estimating function values****16.7.2 Evaluating limits****16.7.3 Approximating integrals****16.7.4 Solving differential equations by series****16.7.5 Error bounds in applications**

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16.8 Uniform convergence preview**16.8.1 Pointwise convergence of functions****16.8.2 Uniform convergence as one error bound for all points****16.8.3 Interchanging limits and integrals****16.8.4 Why power series are trustworthy inside their radius**

Counterexample

16.9 Warning examples

Optional

16.9.1 Taylor series converges but not to the function**16.9.2 Endpoint behavior differs from interior behavior**

Optional

16.9.3 Termwise operations fail for arbitrary function series

16.9.4 A good local approximation becomes bad far away**16.10 Chapter review and projects****16.10.1 Skills: power series and Taylor series****16.10.2 Project: build a sine calculator with error control****16.10.3 Project: compare Taylor approximations visually****16.10.4 Hook: the series for sine and cosine hide a rotating vector**Computing /
ProjectComputing /
Project

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Capstones and Bridges

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17.1 Complex numbers and Euler's formula

Bridge / Optional

17.1.1 Complex plane

17.1.2 Polar form

17.1.3 Multiplication as rotation and scaling

17.1.4 Euler's formula from Taylor series

17.1.5 Why e^{it} is the natural language of oscillation

17.2 Vibrations and second-order equations

Optional

17.2.1 The spring equation

17.2.2 Sine and cosine as natural motions

17.2.3 Energy conservation

17.2.4 Damping and forcing preview

17.3 Fourier series preview

Optional

17.3.1 Building a periodic function from waves

17.3.2 Orthogonality of sine and cosine, geometrically

17.3.3 Square waves and Gibbs phenomenon

17.3.4 Why Fourier series belong after Taylor series

17.4 The one-dimensional Stokes theorem

Bridge

17.4.1 Boundary of an interval

17.4.2 $\int_a^b F'(x) dx = F(b) - F(a)$

17.4.3 The FTC as a boundary theorem**17.4.4 Preview of line integrals and differential forms**

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17.5 Conservation laws in one dimension**17.5.1 Density and flux on a line****17.5.2 Accumulation inside an interval****17.5.3 What crosses the boundary changes what is inside****17.5.4 Preview of divergence in multivariable calculus**

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17.6 Least action and the shape of a path**17.6.1 From optimization of numbers to optimization of functions****17.6.2 The brachistochrone story****17.6.3 A gentle first variation calculation****17.6.4 Why this is beyond ordinary single-variable calculus****17.7 Final projects**

Project

17.7.1 Model, solve, and test a cooling law

Project

17.7.2 Approximate a difficult integral three waysComputing /
Project**17.7.3 Build a Taylor-series calculator**

Project

17.7.4 Analyze a parametric curve from motion data

Bridge / Project

17.7.5 Explain the FTC as the first boundary theorem

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- D.1 Least upper bound property**
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